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1 The Directive

In October 2026 the fleet's principal issued wave-66 with a single, dense instruction: go further than the waves before it; study the other works pushed recently to SuperInstance, because they can help the fleet go much further; dog-food the team on what others have done, both inside and outside the instance, and do it slowly over many rounds; decompose ideas into elementary parts in spreadsheet logic with the help of jevs finding gates through the layers of logic as the works grow and evolve; keep timestamped ledgers so the fleet understands more and more as the works break down how they operate; and level an EXOJ of how to do it again without the agents. This white paper is the closing receipt of that instruction, and every claim in it carries a pointer into the ledgers where the proof lives.

The instruction contains four distinct demands that previous waves had only partially addressed. First, the corpus of study was to be the fleet's own sibling works — the twenty-nine repositories and research streams that SuperInstance has pushed — rather than fresh territory. Second, the decomposition had to land in spreadsheet logic: elementary parts as rows, attributes as columns, gates and failure modes as cells, so that any reader can sort, filter, and audit the fleet's machinery the way an auditor reads a ledger. Third, judgment itself was to be delegated to the jev machinery the fleet had already built — soft, non-collapsing evaluation with explicit observation reserved for holes. Fourth, and most unusually, the wave had to finish by destroying its own dependency on agents: the method must survive as an executable that a fresh instance can run with no subagent fleet at all.

The order of operations followed the instruction's own cadence — slowly, over many rounds. Round one dispatched six parallel lanes across six families of works; round two swept the resulting decompositions with the jev gate machinery; round three compiled the spreadsheet atlas; round four leveled the EXOJ kit inside the exoj repository; round five produced this document. Each round appended timestamped receipts to append-only ledgers before the next round began, so the wave's own history is auditable in the same style as the works it studied. Nothing in this paper rests on memory: every number is re-derivable from the committed artifacts.

29	528	401	127
WORKS STUDIED	ELEMENTARY PARTS	GATES SWEEPED	HOLES OBSERVED

2 The Corpus: Twenty-Nine Works

Wave-66 studied every substantial work the instance had pushed, organized into six families that reflect how the fleet actually divides its concerns. The substrate family holds the representation layer — qthe's one-byte ternary hyper-embedding, quilt-jepa's predictive world model in a cell mesh, jev-quilt's typed decision-surface cells, and quilt-qcells' receipt ledgers where the ledger itself is the quantum circuit. The garden family holds the growth machinery: the living jev-garden training system, the jeviter weave lifecycle, exoj's non-collapsing reasoning surface, and the fleet-seeds seedbox that spawns charter-bound repositories. The swarm family carries multi-agent trust: quilt-murmur's gossip economy, the quilt-pincher reflex organism, and crab-traps' tamper-evident lures.

The organ family is the fleet's evidence infrastructure — three live Cloudflare workers, the quilt-jev-toolkit organ protocol with signed checkpoints and partial custody, the MCP receipt chain that exposes the fleet's ledgers to any agent on the account, and the research canons that bind sprint lineage to pre-registered predictions. The labs family drives the substrate through real domains: a quant trading lab, a database-style vault lane with exact rewind, raw quilt primitives, a chain-of-thought pipeline, two moth quantum-randomness efforts, and supporting research. The meta family completes the corpus with the account's public site, its fleet infrastructure, and — deliberately — the works outside the instance that shaped it: the upstream quilt repository with its hundred-essay corpus, and five external research streams from V-JEPA 2 to hyperdimensional computing that the fleet's scouts had receipted months earlier.

family	works	what it holds
substrate	4	qthe, quilt-jepa, jev-quilt, quilt-qcells — the representation layer judgment stands on
garden	4	jev-garden, jeviter, exoj, fleet-seeds — growth, lifecycle, idle-time compilation
swarm	3	quilt-murmur, quilt-pincher, crab-traps — trust, gossip, reflexes, tamper traps
organ	4	organ workers, jev-toolkit, mcp-receipts, research-canons — receipts and tamper evidence
labs	6	quant, dba, raw, cot-quilt, MicroMoth, moth-research — applied domains driving the substrate
meta	8	site, si-fleet, atlas, codespace, prospector, craftmind, upstream essays, external scouts

Table 1 — The wave-66 corpus: 29 works across 6 families.

Dog-fooding was mandatory before decomposition: each lane ran the cheapest honest verification a work offers — its smoke test or selftest — under a strict no-network, no-key, under-ninety-seconds law. Most works passed under their own discipline: jev-garden 30/30, jeviter 85/85, quilt-quant 17/17 offline, quilt-jev-toolkit 44/44, the qcells tamper sweep 4254/4254. The verdicts that were not green were receipted rather than hidden, and several of those honest negatives became findings in their own right — a workspace whose normalized file timestamps break mtime-bound seals, a repo root smoke broken by a moved export while its canonical experiment stays green, an import manifest staled by later receipts doing exactly its job by failing loudly.

3 The Method: Spreadsheet Logic on Five Layers

Every work was decomposed by one binding schema, written before any decomposition began. Each elementary part is a row with a stable identifier, a layer, a one-sentence essence, declared inputs and outputs, a gate, a gate kind, a failure mode, an elementarity flag, and — the law that keeps the atlas honest — a file-and-line evidence pointer into the studied repository that the decomposing lane actually read. Elementary means the part cannot be split further without changing its meaning; if a lane could not evidence a part, the part did not enter the atlas. Alongside parts, each work contributes its two to five core ideas mapped to the parts that express them, and a sheet block re-rendering the work as quilt cells in the fleet's own nine-kind cell vocabulary, so the atlas doubles as a migration sketch into the shared substrate.

The five canonical layers are the wave's answer to the principal's phrase 'layers of logic'. They order every work's internals from the ground up, and because the ordering is shared across all twenty-nine works, the atlas can compare a quantum ledger's policy layer with a gossip economy's policy layer directly. A gate is any pass-or-fail condition a part asserts before, during, or after it acts; the vocabulary is fixed so the gate sweep can treat a seal in an organ protocol and an admission control in a swarm protocol as the same kind of thing.

layer	name	the question it answers
0	substrate	what irreducible representation does the work stand on?
1	mechanism	what transforms state, and from what to what?
2	policy	what decides, admits, prices, or refuses?
3	interface	what does the outside touch — wires, tools, sheets?
4	evidence	what proves it ran, and what makes tampering loud?

Table 2 — The five canonical layers of logic.

Two doctrinal rules governed every lane. Decision rules came before runs: the sweep's verdict rules were receipted into the ledger before the sweep executed, never after. And honest negatives were crown jewels: a gate the lanes could not find, a test that could not run, a mechanism the docs oversell — all of these were recorded as first-class rows, because the wave's purpose was not to flatter the fleet's output but to understand it part by part.

4 The Gate Sweep: the Jevs Walk the Layers

With five hundred twenty-eight parts decomposed, the wave let the jevs do what the principal asked of them: find the gates through the layers of logic. The sweep is an ExoJ field — the fleet's non-collapsing reasoning surface, running the ledger conservation policy that earlier experiments established as the naturality fix. Each work occupies a column of a hex lattice; its five layers are rows. Walking every work from layer zero to layer four, the sweep reads each part's gate and applies six rules that were receipted before the run began:

R1 — a part with a non-empty gate and a known gate kind is OPEN, and receives a soft deformation of gamma 0.75, eta 0.25, delta 0.50: high confidence mass, low possibility mass, creativity held in the creative band. R2 — a gate with no kind is SOFT, judged at 0.50/0.50/0.50. R3 — a part with no gate at all is a HOLE, and the only collapse in the entire system happens there: an explicit, recorded, local observation at delta 0.20. R4 — seal and conservation gates receive a second reinforcing emission. R5 — a work missing an entire layer is itself a hole, observed at that layer's cell. R6 — a work with fewer than five gated parts earns a thinness warning. The field is the proof object: 606 content-addressed links, verified from genesis.

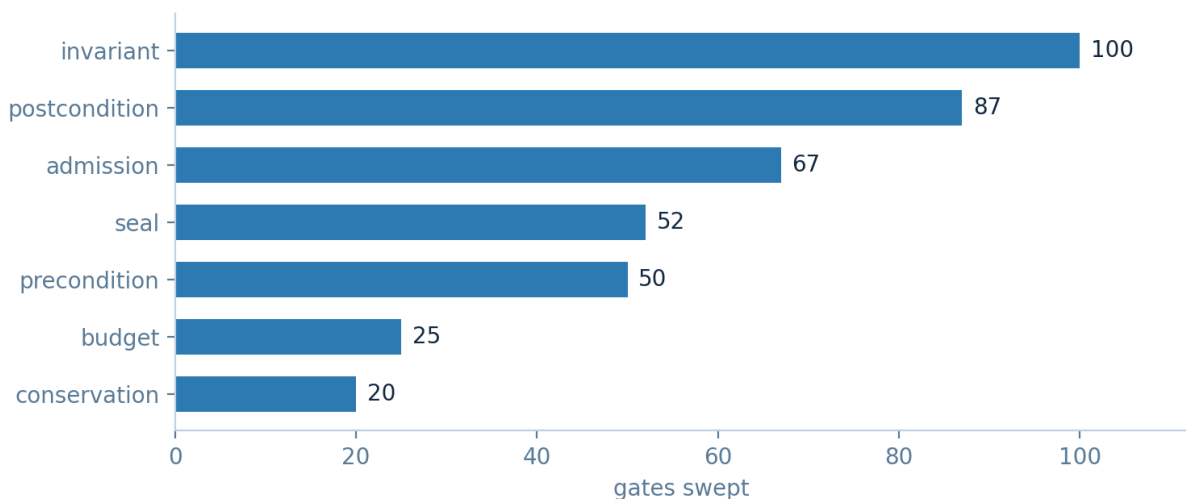


Figure 1 — Gate kinds across the atlas: 401 swept gates by kind.

The verdict: 401 of 528 parts carry concrete gates — 75.9 percent of the fleet's decomposed logic is guarded — and every one of them resolved OPEN, meaning no lane wrote a gate it could not classify. The 127 holes are the wave's most useful yield: unguarded parts are not defects to hide but the next round's work queue, and they concentrate exactly where the fleet's documentation lives rather than where its mechanisms live. Invariants dominate the gate-kind histogram at one hundred, followed by postconditions and admission controls; conservation laws and budgets are fewer but load-bearing, appearing at the substrate and policy layers of the most tamper-resistant works.

Because the field never collapses except at holes, each work closes with a `prob_open` reading — the fraction of its judged logic still held open. That number is a live meter of how much of a work remains unobserved after the sweep, and it is recomputed byte-identically by the kit in chapter six. The field itself is dunnable:

saved as an ExoJ shell, reloadable, re-verifiable, and content-addressed end to end, so the wave's judgment is itself part of the evidence layer it studies.

5 Findings

The atlas's first finding is a convergence: across six families that grew largely independently, the same four moves recur at the same layers. Every family independently arrived at an irreducible typed row or cell as its substrate; every family hash-chains its evidence append-only; every family refuses fail-closed with named errors rather than failing silent; and every family preserves its honest failures — a recorded erratum, a refuted ensemble, an unsealed-opcode hole — as load-bearing parts of its evidence layer. The fleet did not decree this convergence; the atlas is the first instrument able to see it, because the atlas is the first artifact that renders all twenty-nine works in one schema.

family	parts	gates	holes	gated %	reading
substrate	81	74	7	91%	representation guarded to the bit; bijections and bounds sealed
garden	92	89	3	97%	lifecycle law: promotion, discard, and observation all gated
swarm	68	49	19	72%	economics guarded; reflex wiring carries the holes
organ	75	69	6	92%	the tamper-evidence core: 26 named fail-closed codes in one work alone
labs	78	65	13	83%	applied discipline holds under real domains; docs lag mechanisms
meta	134	55	79	41%	prose and positioning: the logic lives in text, not gates — the wave's queue

Table 3 — Sweep results by family: parts, gated parts, observed holes.

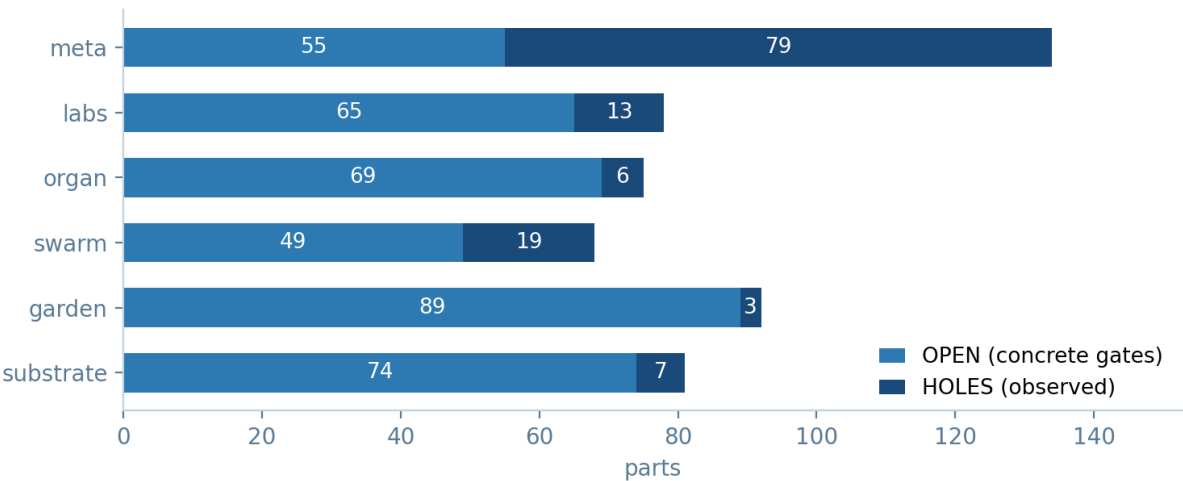


Figure 2 — OPEN gates versus observed holes per family; the meta family is the work queue.

The second finding is the shape of the hole distribution. The organ and garden families are gate-dense — the organ family's toolkit alone carries a twenty-six-code fail-closed law with file-and-line evidence for every code — while the meta family is hole-dense, at forty-one percent gated. That is not an indictment: a positioning site and an essay corpus are prose, and prose resists gates by nature. But it tells the fleet precisely where the next wave's decomposition effort pays: turning the account's public claims into testable, gatable parts, or consciously marking them as out of scope.

The third finding is operational. The dog-food round surfaced environment-level traps that no single repo could have seen alone: workspace file mtimes normalized to one value, which breaks every mtime-bound seal the fleet uses and argues for content-hash seals going forward; a node-version quirk where an explicit test-directory invocation fails while package-script invocation passes; two repos whose own scripts rewrite their output artifacts mid-run, which the lanes handled by restoring committed state and receipting the incident. Each of these was receipted with its evidence, and each is a candidate gate for a future fleet-standard smoke harness.

6 The EXOJ Kit: Doing It Again Without Them

The wave's final demand was the uncommon one: level an EXOJ of how to do it again without the agents. The exoj repository now carries the atlas kit — atlas.mjs, a zero-dependency executable with four verbs. Verify re-checks the bundled artifact of record in six checks, from corpus integrity through referential integrity of every idea-to-part and cell-to-part reference to the 606-link chain of the saved judgment field. Sweep re-runs the gate walk from the pre-registered rules, which live in the module itself so they can never silently drift. Csv emits the whole spreadsheet logic — corpus, parts, gates, ideas, cells — for any spreadsheet tool. Protocol prints the seven-move runbook, from registry to seal, for a fresh instance with no fleet.

```
node atlas.mjs verify # artifact of record: 6 checks
node atlas.mjs sweep # pre-registered gate walk, exoj policy=ledger
node atlas.mjs csv # corpus / parts / gates / ideas / cells as CSV
node atlas.mjs protocol # the seven moves, start to seal
node experiments/e_x8_atlas_gatesweep.mjs # replay == artifact of record
```

The claim that the kit reproduces the wave is itself receipted as experiment E-X8, and its result is the strongest number in this paper: the kit's replay lands on the byte-identical chain tip as the live sweep — 13463fd0, six hundred six links — with all twenty-nine per-work verdict maps and both histograms exactly equal, and a second replay run proving determinism. Offline, keyless, agent-free, the wave re-runs itself from its own bones. The experiment also caught its own harness bug during development — the kit's command-line interface executed on import under ESM, silently ending the replay with a usage message — which is exactly the class of failure the kit exists to make loud rather than silent.

E-X8 check	result
R2 kit self-verification	PASS — 6/6 (corpus, parts, refs, artifact totals, field chain, ledgers)
R1 replay totals	PASS — parts 528 / gated 401 / OPEN 401 / SOFT 0 / HOLE 127, equal
R1 per-work verdict maps	PASS — 29/29 works agree on OPEN/SOFT/HOLE and seal reinforcement
R1 histograms	PASS — gate-kind and layer histograms identical
R1 chain tip	PASS — byte-identical 13463fd0... (606 links)
R3 determinism	PASS — two kit runs land on the same chain tip

Table 4 — E-X8: the kit replays the wave without agents.

move	act	leaves behind
1	REGISTRY — corpus.json of every work worth studying, in and out of the instance	the map
2	DECOMPOSE — elementary parts per the binding schema, smoke-verdict first	parts / family / work.json
3	PREREGISTER — sweep rules receipted before any judgment	rules row, chained
4	SWEEP — the jev walk: soft emits for gates, observation at holes	gate-map + field shell
5	READ THE MAP — gated percentage, hole queue, prob_open per work	the next round's queue
6	COMPILE — spreadsheet logic for humans, with live cross-checks	atlas workbook / CSVs
7	SEAL — verify, commit, append-only, hand the bones to the next instance	the kit itself

Table 5 — The seven moves of the decomposition protocol.

One honesty note belongs in the closing chapter rather than a footnote: this container holds no GitHub credential material, so the kit's commit — a993ed6, the artifact of record — is verified local only, and the push is receipted as pending rather than claimed. The fleet's push discipline requires remote-equals-local verification, and that step will complete wherever a token exists; nothing about the kit or the atlas depends on it. The deeper point the wave leaves behind is the one the principal asked for: the method now has an exoj. The next instance does not inherit a story about how wave-66 was made — it inherits an executable that makes wave-66 again, and a ledger that proves the two are indistinguishable.